

Abstract

This research paper proposes a machine learning model for the detection of four different neurological states - physical stress, cognitive stress, emotional stress and relaxation through focusing on non-EEG physiological signals whose data was collected from 20 healthy subjects using wrist worn biosensors and consists of electrodermal activity(EDA), temperature, acceleration, heart rate (HR) and arterial oxygen level(SpO₂). We evaluated the data in both supervised and semi supervised frameworks.

In supervised learning, we used different models out of which XGBoost model to train on the dataset after standard data preprocessing and cleaning that had given 98% accuracy which is the best attained so far.

In semi-supervised learning, we had done feature engineering, divided the data in 30 sec window and in this XGBoost attained 96% accuracy, Random Forest reached 95% accuracy.

Our report highlights that supervised models had performed the best with fully labeled data, whereas semi supervised methods had also given strong performance in partially labelled data.

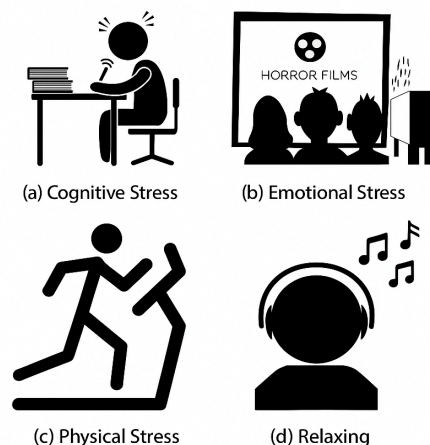
KEYWORDS

Supervised learning, semi supervised learning, Label Spreading, physiological signals, stress classification, XGBoost.

I. INTRODUCTION

Keeping track of neurological states like stress, emotions, and mental workload is really important—whether it's for mental health, improving how we interact with tech, or creating more personalized care. EEG is great for directly reading brain activity, but in real life, it's kind of a hassle because it's expensive and not exactly user-friendly for daily use. Thanks to wearables, we now have easier ways to gather data through things like heart rate, EDA, SpO₂, temperature, and movement. These signals can still reflect what's going on in the brain without needing complex equipment.

In this study, we're using non-EEG signals along with machine learning—both supervised and semi-supervised approaches—to predict neurological states. The goal is to move away from relying on EEG and create something that's more practical, scalable, and useful for real-time monitoring in everyday life.



Several recent studies have shown that it's possible to estimate neurological states using signals other than EEG. For example, one study [1] used a motif-based method to

analyze patterns in biosignals like skin conductance, temperature, and pulse rate. While the results were promising, their approach mainly worked with short signal segments and struggled with noisy or real-world data that changes over time.

Ahmadian et al. [2] explored non-invasive EEG for predicting movement using brain signals. Unlike our wearable-based approach, EEG requires complex setup and is less suited for everyday, real-world use.

Polhemus et al. [3] conducted a systematic review on how individuals with chronic neurological and mental health conditions perceive data visualizations for self-management. While it highlights the value of feedback, it doesn't address how such physiological data can be automatically classified using machine learning for real-time state detection, which our work aims to achieve. Neurologic Wake-Up Tests (NWTs) [4] are the current gold standard for assessing sedated brain-injured ICU patients, relying on momentary sedation interruption to evaluate neurological status. NWTs are invasive, stress-inducing, and limited in frequency—unlike our approach, which enables continuous, non-invasive monitoring through wearable physiological signals. This review [5] highlights the growing potential of wearable physiological data and machine learning models for real-time stress prediction and personalized management. While it emphasizes stress detection broadly, it lacks a focused, multi-state classification approach like ours, which targets specific neurological states beyond just stress. This study [6] proposed a highly accurate EEG-based ML system for detecting and classifying mental stress using hybrid features and reduced electrode setups. The system depends on EEG signals, which require electrode placement and are less practical for everyday, non-clinical use. This paper [7] reviews current advancements in smart wearable systems for continuous, non-invasive health and mental

status monitoring. While it provides a broad overview of wearable systems for health monitoring, it lacks focus on specific machine learning methods for classifying neurological or mental states. The review [8] compares various emotion recognition methods—including facial analysis, EEG, and smart wearables—highlighting their suitability across different human-computer interaction applications. It provides a broad overview but lacks detailed implementation strategies for real-time, wearable-based emotion recognition using physiological signals alone. [9] study developed a user-independent emotion recognition system using ECG, skin temperature, and EDA, achieving up to 78.4% accuracy with SVM classification. Despite using wearable-friendly signals, the system faced significant classification challenges due to high intra-class variance and overlapping feature distributions. [10] applied tensor decomposition on non-EEG physiological data to visualize and classify neurological stress states with up to 100% accuracy. While promising, the method focuses mainly on visualization and lacks real-time or deployable model integration for continuous monitoring. The Nonin WristOx2 3150 [11] is a compact, wrist-worn pulse oximeter designed to non-invasively monitor arterial oxygen saturation (SpO_2) and pulse rate. While reliable for oxygen and heart rate monitoring, it does not capture broader physiological signals needed for comprehensive neurological state classification. [12] reviews various sensor-based approaches—contactless, contact, and invasive—for automated emotion recognition and classifies them by use case and expected outcomes. While it categorizes sensor technologies, it lacks focus on integrated machine learning models for real-time, wearable, and non-invasive emotion recognition. This [13] study used EEG, EMG, and PPG signals with wavelet transform and artificial neural networks to predict epileptic seizures before onset. Its reliance on EEG and high-complexity ANN architectures poses challenges for real-time, low-power wearable

implementation.[14] paper presents a comprehensive review of emotion recognition using physiological signals, covering datasets, features, classifiers, and system frameworks.While extensive, the review primarily summarizes existing work and lacks implementation insights for real-time, wearable, multi-state emotion recognition systems.

Our work builds on these ideas by expanding beyond just stress to include cognitive, emotional, and physical states. We also introduce a semi-supervised learning approach that makes better use of partially labeled data, and we apply a robust 30-second windowing strategy to capture meaningful patterns while filtering out noise.

In this study, we propose a practical and fully non-EEG-based framework for classifying neurological states using only non-invasive physiological signals—namely SpO₂, heart rate (HR), electrodermal activity (EDA), skin temperature, and accelerometer data. Our key contributions are:

1. **Non-Invasive, EEG-Free Design**
We designed a complete pipeline that avoids EEG entirely, relying instead on physiological signals that can be captured easily with wearables. This makes our approach more suitable for real-world, continuous monitoring.
2. **Supervised Learning with High Accuracy**
We cleaned and preprocessed the data by removing outliers and handling missing values, then extracted features from synchronized, windowed segments. Among the models tested, XGBoost achieved the best

performance with **98% accuracy**, showing strong potential for state classification using only peripheral signals.

3. **Semi-Supervised Learning for Low-Label Settings**

To simulate real-world conditions where labels are limited, we used a semi-supervised approach. We segmented the data into 30-second windows to reduce noise and applied **Label Spreading** to infer pseudo-labels from a small labeled set. These were combined with the original labels to train models—XGBoost and Random Forest achieved **96% and 95% accuracy**, respectively.

4. **Windowing for Robust Signal Processing**

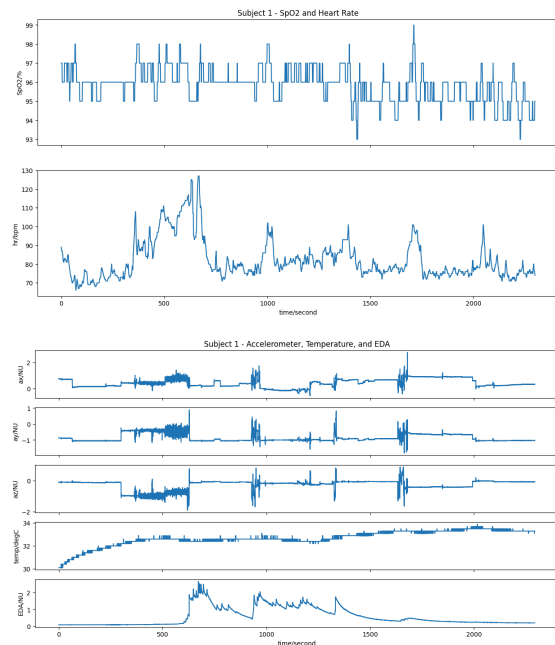
Using fixed 30-second windows helped reduce random fluctuations and improved model stability, especially for noisy signals like EDA and motion data.

Overall, our framework demonstrates that reliable neurological state prediction is possible using only wearable, non-invasive sensors—offering a more accessible and scalable alternative to EEG-based systems.

II. Preliminaries and Dataset Description

This database contains non-EEG physiological signals collected at Quality of Life Laboratory at University of Texas at Dallas, used to infer the neurological status (including physical stress, cognitive stress, emotional stress and relaxation) of

20 healthy subjects. The data was collected using non-invasive wrist worn biosensors and consists of electrodermal activity (EDA), temperature, acceleration, heart rate (HR), and arterial oxygen level (SpO2).



The experimental procedures involving human subjects described in this work were approved under UTD IRB # 12-29 by the Institutional Review Board at the University of Texas at Dallas, Richardson, Texas, USA.

The dataset consists of 7 stages for 20 subjects:

1. First relaxation: 5 minutes
2. Physical Stress: Stand for one minute, walk on a treadmill at one mile per hour for two minutes, then walk/jog on the treadmill at three miles per hour for two minutes
3. Second Relaxation: five minutes.
4. Stress:
 1. Mini-emotional stress*: 40 seconds
(Note: This portion of the data,

which was collected right before the cognitive stress task, is not explained in the paper.) During this 40 seconds, the “instructions” for the math portion of the cognitive stress (to count backwards by sevens, beginning with 2485, for three minutes) were read to the volunteer.

2. Cognitive Stress: Count backwards by sevens, beginning with 2485, for three minutes. Next, perform the Stroop test for two minutes. The volunteer was alerted to errors by a buzzer. The Stroop test consisted of reading the names of colors written in a different color ink, then saying what color the ink was.

5. Third Relaxation: five minutes.

6. Emotional Stress: The volunteer was told he/she would be shown a five minute clip from a horror movie in one minute. After the minute of anticipation, a clip from a zombie apocalypse movie, The Horde was shown.

7. Forth Relaxation: five minutes.

Files:

The data files are provided in WFDB format with two records per subject: one that contains the accelerometer, temperature, and EDA signals, and one that contains the SpO2 and heart rate signals. Header files also contain information about the subject. There is one annotation file per subject that indicates the time locations and labels of the transition states. The subjectinfo.csv file

also contains information about each subject.

III. Proposed Methodology

A.Dataset Preprocessing and Analysis

1. Data Acquisition and Structure

Raw physiological signals were obtained using the **wfdb** Python package, which reads **.dat**, **.hea**, and **.atr** files. Each subject had two separate recordings:

- 1. **SpO2HR**: Contains Heart Rate and Oxygen Saturation at 1Hz.
- 2. **AccTempEDA**: Contains 3-axis Accelerometer, Temperature, and EDA at 8Hz.

2. Temporal Alignment and Downsampling

To integrate these two different frequency data :

- Accelerometer, EDA, and Temperature (8Hz) were **downsampled to 1Hz** by selecting every 8th sample.

3. Annotation Integration

Task states (e.g., Relax, EmotionalStress) were extracted from **.atr** annotation files and mapped to each extracted datapoint.

4.Demographic Merging

Subject specific demographic data(height, weight, gender) were joined in accordance with the subject, from a csv file containing the

information about the all 20 subjects, creating the fully integrated final dataset.

5. Label Encoding

Annotations were mapped to numeric values to facilitate model training:

State	Label
Emotional Stress	0
Relax	1
Physical Stress	2
Cognitive Stress	3
Unknown	-1

Unmappable or ambiguous labels were filtered or flagged for further handling.

Purpose and Justification:

This step ensured that raw, high-frequency sensor data were transformed into a structured, labeled, and demographically aware dataset suitable for downstream machine learning tasks. The time-aligned annotations reflect real-world neurological states, enabling supervised learning without EEG.

for subject in all_subjects:

```
    load SpO2HR and AccTempEDA signals
    downsample AccTempEDA (8Hz → 1Hz)
    truncate all signals to min length

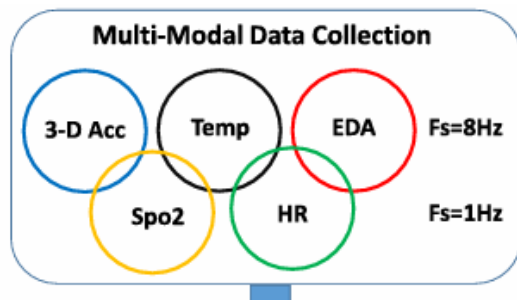
    extract .atr annotations and map to 1Hz timeline

    load demographic metadata from CSV
```

construct DataFrame with signals +
annotation + demographics

append to master dataset

save final merged dataset as CSV



B. Methodology — Supervised Learning Pipeline

The supervised learning pipeline is used to classify neurological stress states using annotated and preprocessed dataset described earlier. The entire process covers exploratory data analysis (EDA), feature engineering, outlier handling, scaling, and model training and prediction using XGBoost.

1. Exploratory Data Analysis (EDA)

Initial analysis involved visual and statistical summary of the dataset:

- Checked for **missing values** (absolute and percentage) across all features.
- Detected and removed **duplicate rows**.
- Investigated the distribution of target labels (**Annotation**, **Annotation_Numeric**).

- Generated **distribution plots**, **boxplots**, and **correlation heatmaps** to identify relationships between features and target variables.
- Plotted time series of physiological signals to observe how they changed during stress-related tasks.

2. Outlier Detection and Correction

To identify and handle anomalous values we had done:

- **Z-score-based detection** for global anomaly count estimation.
- **IQR-based outlier handling**, categorized as:
 - **Strict capping per subject** for SpO₂, HR, and Temp.
 - **Global threshold capping** ($3 \times \text{IQR}$ for sensors like accelerometer and EDA).
- Flagged rows with 3 or more outliers as suspicious but kept them after cleaning to test model robustness.

After removing outliers, distributions were visualized again to confirm cleaned patterns using boxplots and KDE histograms.

3. Feature Scaling

Standardization was applied to all continuous features:

- Features were transformed using **StandardScaler** to ensure mean = 0 and std = 1.
- Original categorical and labels (Gender, Annotation) were preserved and were reattached after scaling the other features.

4. Model Training

The XGBoost classifier was used as the main model for supervised learning:

- Focused mainly on this algorithm for its performance on structured tabular data.
- We had trained the model on the fully labeled , scaled dataset .
- Final model was evaluated using classification metrics such as accuracy, precision, recall, and F1-score.

Purpose and Justification:

Supervised learning was used as a baseline to test if physiological signals can classify neurological states. XGBoost was selected for its strength with imbalanced, complex and high-dimensional data. Preprocessing steps such as outlier handling and scaling were necessary to ensure better model performance and reduction of the sensor noise.

Pseudo Code:

Require: Dataset $D = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$

where $x_i \in \mathbb{R}^d, y_i \in \mathbb{R}$

Ensure: Trained model f , Evaluation metrics (MSE, Accuracy)

1: Dataset Definition ▷ Define the dataset

2: Missing Value Handling

3: for all $x_i \square = \text{null}$ do

4: Handle by imputation or deletion

5: end for

6: Descriptive Statistics

$$\mu \square = 1/n \sum_{i=1}^n x_i \square$$

$$\sigma^2 \square = 1/n \sum_{i=1}^n (x_i \square - \mu \square)^2$$

7: Duplicate Removal

$$D' = \{(x_i, y_i) \in D \mid (x_i, y_i) \notin D_{i \neq 1} \square \}$$

8: Outlier Detection

$$\text{Z-score: } z_i \square = (x_i \square - \mu \square) / \sigma \square$$

9: if $|z_i \square| > \theta$ then

10: Mark as outlier

11: end if

$$12: \text{IQR: } IQR \square = Q_3 \square - Q_1 \square$$

13: if $x_i \square < Q_1 \square - 1.5 \cdot IQR \square$ or $x_i \square > Q_3 \square + 1.5 \cdot IQR \square$ then

14: Mark as outlier

15: end if

16: Standard Scaling

$$x_i^{\text{scaled}} \square = (x_i \square - \mu \square) / \sigma \square$$

16: Feature Concatenation

$X = [X^{\text{scaled}} \mid C]$ where C = categorical + label features

17: Model Training (XGBoost)

$$f = \arg \min_f \sum_{i=1}^n L(f(x_i), y_i) + \Omega(f)$$

18: Evaluation Metrics

$$\text{MSE: } 1/n \sum_{i=1}^n (f(x_i) - y_i)^2$$

$$\text{Accuracy: } 1/n \sum_{i=1}^n 1 \{f(x_i) = y_i\}$$

C. Methodology — Semi-Supervised Learning Framework

To reflect real-world scenarios with limited labeled data, we applied a semi-supervised

learning approach. Only half of the dataset was labeled, while the rest remained unlabeled, focusing on situations where manual labeling is time-consuming or requires expert knowledge.

1. Partial Labeling and Dataset Preparation

We used the cleaned physiological dataset, we separated the data into two parts:

- **Labeled Set:** 50% of the data with known class labels.
- **Unlabeled Set:** 50% with labels removed (-1), treated as unknown.
- Each data point consisted of **window-averaged physiological features**, including:

SpO₂_mean, HR_mean, Temp_mean,
EDA_mean, Acc_magnitude_mean.

These features were extracted using 30-second non-overlapping windows to summarize time-series behavior. All features were then standardized using **StandardScaler** to ensure uniformity across distributions.

2. Label Spreading for Pseudo-Label Generation

To find out the labels for the unlabeled portion, we applied Label Spreading which is a semi-supervised algorithm, with an RBF kernel and $\alpha = 0.2$. The model assigned labels to the unlabeled data by learning from the labeled data based on feature similarities.

After propagation:

- **Predicted labels** (**transduction_**) were extracted.
- **Label confidence scores** were computed using **label_distributions_**.

- Only **high-confidence predictions** (**confidence > 0.9**) were selected to be used in final training.

This approach increased the size of the labeled training set without manual annotation, while maintaining model reliability.

3. Model Training and Evaluation

The resulting dataset (true labels + high-confidence pseudo-labels) was used to train and compare three classifiers:

a. Logistic Regression

This was selected as a baseline model for its simplicity and interpretability.

It helps assess whether basic linear patterns in the features are enough to separate stress states.

- **Accuracy:** ~93%
- **Strengths:** Quick to train, easy to understand, and less prone to overfitting.
- **Limitation:** Struggles with complex or non-linear patterns in the data.

b. Random Forest Classifier

This ensemble method builds multiple decision trees and combines their predictions.

It works well for capturing non-linear relationships in physiological data.

- **Accuracy:** ~95%
- **Strengths:** Handles feature interactions well and are robust to noise.
- **Limitation:** Larger model size and less interpretability compared to logistic regression.

c. XGBoost Classifier

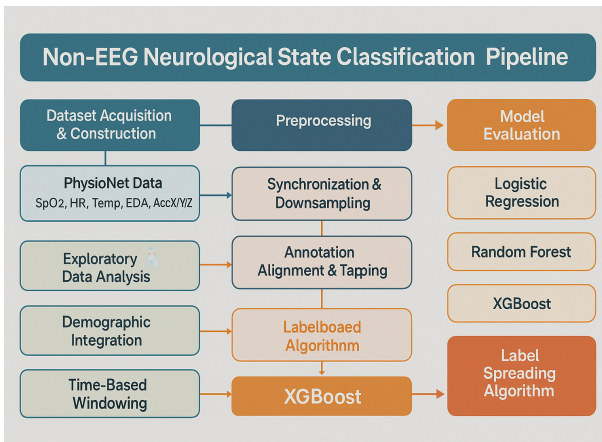
XGBoost is a powerful, gradient-boosted tree-based model designed for high accuracy

and speed.
It delivered the best results among all models.

- **Accuracy:** ~96%
- **Strengths:** Excellent accuracy, effective on imbalanced datasets, and includes built-in regularization.
- **Limitation:** Slower to train and requires careful hyperparameter tuning.

Purpose and Outcome

This semi-supervised approach effectively leveraged partially labeled physiological data by generating labels for the unlabeled portion using model confidence. The findings show that even with limited annotations, high-performing models can be developed through pseudo-labeling. Among all models evaluated, XGBoost consistently delivered the best results, highlighting its strength in capturing complex physiological patterns linked to neurological states.



IV. Results

Semi-Supervised Learning

Model	Acc urac y	Macro Precisio n	Macro Recall	Macro F1-scor e
Logistic	0.68	0.66	0.56	0.59
Random Forest	0.95	0.95	0.94	0.94
XGBoost	0.96	0.96	0.96	0.96

Semi Supervised XGBoost Model
claasification report

	precision	recall	F1_sc ore	support
0	0.92	0.93	0.92	95
1	0.97	0.98	0.98	301
2	0.99	0.95	0.97	83
3	0.97	0.97	0.97	89
accura cy			0.96	568
Macro avg	0.96	0.96	0.96	568
Weight ed avg	0.96	0.96	0.96	568

Supervised XG Boost classification report

	precision	recall	F1_scor e	support
0	0.97	0.94	0.95	1475
1	0.98	0.99	0.98	4008
2	0.99	0.99	0.99	1300
3	0.97	0.97	0.97	1375
accuracy			0.98	8158
Macro avg	0.98	0.97	0.97	8158
Weighted avg	0.98	0.98	0.98	8158

V. Conclusion

This study demonstrates that neurological states can be accurately classified using only physiological signals—without relying on EEG—by applying structured machine learning methods. We implemented both supervised and semi-supervised learning pipelines on the PhysioNet Non-EEG dataset, which comprises multimodal signals including SpO₂, heart rate, electrodermal activity, skin temperature, and 3-axis accelerometry collected from 20 subjects exposed to stress and relaxation conditions.

In the supervised pipeline, we conducted extensive preprocessing, including missing value analysis, robust outlier detection (Z-score and IQR-based capping), and standard scaling. The application of XGBoost to the fully labeled and cleaned dataset yielded a high classification accuracy of 98%, demonstrating its strength in handling structured physiological features.

In the semi-supervised learning setup, we simulated real-world conditions by using only 50% of the data for initial labeling. Label Spreading was applied to infer the missing labels based on feature similarity, followed by confidence-based filtering. A 30-second window-based feature extraction strategy was used to preserve temporal structure while minimizing signal noise. The resulting augmented dataset—enhanced with high-confidence pseudo-labels—was used to train three classifiers: Logistic Regression (93% accuracy), Random Forest (95%), and XGBoost (96%). These results demonstrate that strong classification performance can be achieved even with limited labeled data.

This project highlights the importance of extracting time-based features, using windowing, and cleaning signals properly when working with physiological data. It also shows that combining supervised and

semi-supervised learning helps make better use of limited labeled data. Overall, the results support the use of non-invasive wearable sensors for real-time stress and relaxation detection, with potential applications in mental health, human-computer interaction, and personalized healthcare.

References

1. Kaya, Y., & Ertuğrul, Ö. F. (2019). Estimation of neurological status from non-electroencephalography bio-signals by motif patterns. *Applied Soft Computing*, 83(105609), 105609. <https://doi.org/10.1016/j.asoc.2019.105609>
2. Ahmadian, P., Cagnoni, S., & Ascari, L. (2013). How capable is non-invasive EEG data of predicting the next movement? A mini review. *Frontiers in Human Neuroscience*, 7, 124. <https://doi.org/10.3389/fnhum.2013.00124>
3. Polhemus, A., Novak, J., Majid, S., Simblett, S., Morris, D., Bruce, S., Burke, P., Dockendorf, M. F., Temesi, G., & Wykes, T. (2022). Data visualization for chronic neurological and mental health condition self-management: Systematic review of user perspectives. *JMIR Mental Health*, 9(4), e25249. <https://doi.org/10.2196/25249>
4. Musick, S., & Alberico, A. (2021). Neurologic assessment of the neurocritical care patient. *Frontiers in Neurology*, 12, 588989. <https://doi.org/10.3389/fneur.2021.588989>

5. Lazarou, E., & Exarchos, T. P. (2024). Predicting stress levels using physiological data: Real-time stress prediction models utilizing wearable devices. *AIMS Neuroscience*, 11(2), 76–102.
<https://doi.org/10.3934/Neuroscience.2024006>

6. Attallah, O. (2020). An effective mental stress state detection and evaluation system using minimum number of frontal brain electrodes. *Diagnostics (Basel, Switzerland)*, 10(5), 292.
<https://doi.org/10.3390/diagnostics10050292>

7. Chan, M., Estève, D., Fourniols, J.-Y., Escriba, C., & Campo, E. (2012). Smart wearable systems: current status and future challenges. *Artificial Intelligence in Medicine*, 56(3), 137–156.
<https://doi.org/10.1016/j.artmed.2012.09.003>

8. Egger, M., Ley, M., & Hanke, S. (2019). Emotion recognition from physiological signal analysis: A review. *Electronic Notes in Theoretical Computer Science*, 343, 35–55.
<https://doi.org/10.1016/j.entcs.2019.04.009>

9. Kim, K. H., Bang, S. W., & Kim, S. R. (2004). Emotion recognition system using short-term monitoring of physiological signals. *Medical & Biological Engineering & Computing*, 42(3), 419–427.
<https://doi.org/10.1007/bf02344719>

10. Pham, T. T. T., Déniz, H. R., & Pham, T. D. (2019). Tensor decomposition of non-EEG

physiological signals for visualization and recognition of human stress. *Proceedings of the 2019 11th International Conference on Bioinformatics and Biomedical Technology*.

11. WristOx2® model 3150 OEM with bluetooth® low energy. (2018, October 18). Nonin; Nonin Medical Inc.
<http://www.nonin.com/OEMSolutions/WristOx23150-OEM>

12. (N.d.-a). Mdpi.com. Retrieved June 29, 2025, from
<https://www.mdpi.com/1424-8220/20/3/592>

13. (N.d.-b). Mdpi.com. Retrieved June 29, 2025, from
<https://www.mdpi.com/1424-8220/20/12/3346>

14. (N.d.-c). Mdpi.com. Retrieved June 29, 2025, from
<https://www.mdpi.com/1424-8220/18/7/2074>

